

Elliptic Curve Cryptography: From Mathematical Foundations to Digital Signatures

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Abstract

This report presents a comprehensive analysis of elliptic curve cryptography (ECC), exploring the mathematical foundations of elliptic curves over finite fields and their application to public-key cryptography. We implement point addition and scalar multiplication algorithms, demonstrate the Elliptic Curve Digital Signature Algorithm (ECDSA), and analyze the computational hardness of the Elliptic Curve Discrete Logarithm Problem (ECDLP) that underpins ECC security. Through computational examples using the secp256k1 curve (Bitcoin) and NIST P-256, we illustrate key generation, ECDH key exchange, and digital signature verification with realistic cryptographic parameters.

1 Introduction

Elliptic curve cryptography has revolutionized public-key cryptography since its independent discovery by Neal Koblitz and Victor Miller in 1985. ECC provides equivalent security to RSA with dramatically smaller key sizes: a 256-bit ECC key offers security comparable to a 3072-bit RSA key. This efficiency advantage makes ECC ideal for resource-constrained environments and underlies its adoption in Bitcoin, TLS/SSL, and government cryptographic standards.

The security of ECC rests on the computational intractability of the Elliptic Curve Discrete Logarithm Problem (ECDLP): given points P and $Q = kP$ on an elliptic curve, finding the scalar k is computationally infeasible for properly chosen curves and sufficiently large field sizes. Unlike the integer factorization problem (RSA) or discrete logarithm in finite fields (DSA), no subexponential-time classical algorithm is known for ECDLP.

Definition 1.1 (Elliptic Curve over Finite Field) *An elliptic curve E over a finite field $\mathbb{F}_p$ (where $p > 3$ is prime) is the set of points (x, y) satisfying the Weierstrass equation:*

$$y^2 \equiv x^3 + ax + b \pmod{p} \quad (1)$$

where $a, b \in \mathbb{F}_p$ satisfy $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$ (non-singularity condition), together with a point at infinity $\mathcal{O}$ serving as the identity element.

2 Elliptic Curve Group Law

The points on an elliptic curve form an abelian group under a geometric addition operation. This group structure enables cryptographic protocols by providing a one-way function (scalar multiplication) that is easy to compute but hard to invert.

2.1 Point Addition

Theorem 2.1 (Point Addition Formula) For distinct points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ on elliptic curve $E : y^2 = x^3 + ax + b$ over $\mathbb{F}_p$, the sum $R = P + Q = (x_3, y_3)$ is given by:

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1} \pmod{p} \quad (2)$$

$$x_3 = \lambda^2 - x_1 - x_2 \pmod{p} \quad (3)$$

$$y_3 = \lambda(x_1 - x_3) - y_1 \pmod{p} \quad (4)$$

where division is performed using modular multiplicative inverse.

2.2 Point Doubling

When adding a point to itself ($P = Q$), we use the tangent line:

Theorem 2.2 (Point Doubling Formula) For $P = (x_1, y_1) \neq \mathcal{O}$, the point $R = 2P = (x_3, y_3)$ is computed as:

$$\lambda = \frac{3x_1^2 + a}{2y_1} \pmod{p} \quad (5)$$

$$x_3 = \lambda^2 - 2x_1 \pmod{p} \quad (6)$$

$$y_3 = \lambda(x_1 - x_3) - y_1 \pmod{p} \quad (7)$$

2.3 Scalar Multiplication

Definition 2.1 (Scalar Multiplication) For a point P on elliptic curve E and integer $k > 0$, the scalar multiplication kP is defined as:

$$kP = \underbrace{P + P + \dots + P}_{k \text{ times}} \quad (8)$$

Efficient computation uses the double-and-add algorithm with complexity $O(\log k)$.

3 Cryptographic Curve Standards

Modern cryptographic protocols rely on standardized curves with well-vetted security properties.

Example 3.1 (secp256k1 - Bitcoin Curve) The secp256k1 curve used in Bitcoin is defined by:

$$p = 2^{256} - 2^{32} - 977 \quad (\text{prime field size}) \quad (9)$$

$$a = 0, \quad b = 7 \quad (\text{curve equation: } y^2 = x^3 + 7) \quad (10)$$

$$G = (\text{generator point with order } n) \quad (11)$$

$$n = \text{FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF E BAAEDCE6 AF48A03B BFD25E8C D0364141}_{16} \quad (12)$$

The order n has 256 bits, providing approximately 128-bit security against ECDLP attacks.

Example 3.2 (NIST P-256) The P-256 curve (also known as secp256r1 or prime256v1) is specified in FIPS 186-4:

$$p = 2^{256} - 2^{224} + 2^{192} + 2^{96} - 1 \quad (13)$$

$$a = p - 3 \quad (14)$$

$$b = 5AC635D8 AA3A93E7 B3EBBD55 769886BC 651D06B0 CC53B0F6 3BCE3C3E 27D2604B_{16} \quad (15)$$

P-256 is widely used in TLS, government applications, and Apple's Secure Enclave.

4 Computational Implementation

The following implementation demonstrates elliptic curve operations over finite fields, including point arithmetic, scalar multiplication via double-and-add, and visualization of curve geometry. We begin by implementing the Extended Euclidean Algorithm for computing modular inverses, which are essential for point addition in finite field arithmetic. The core `EllipticCurve` class provides methods for point validation, addition, doubling, and scalar multiplication using the double-and-add algorithm with $O(\log k)$ complexity. We then demonstrate practical cryptographic protocols including ECDH key exchange and ECDSA signature generation and verification using realistic curve parameters.

The computational implementation successfully demonstrates all core cryptographic operations. The small curve over $\mathbb{F}_{97}$ reveals the discrete point structure characteristic of finite field elliptic curves, with the generator point producing a cyclic subgroup under scalar multiplication. Our ECDH key exchange demonstrates how Alice and Bob can establish a shared secret using only public key exchange, with both parties computing identical shared secrets from their respective private keys and the other party's public key. The ECDSA signature implementation correctly validates authentic messages while rejecting tampered messages, demonstrating the cryptographic security property that even single-bit message alterations produce signature verification failures.

5 ECDSA: Elliptic Curve Digital Signature Algorithm

The Elliptic Curve Digital Signature Algorithm provides authentication and non-repudiation through public-key cryptography.

5.1 Key Generation

Definition 5.1 (ECDSA Key Pair) *Given elliptic curve E with generator point G of prime order n :*

1. *Select random private key $d \in [1, n - 1]$*
2. *Compute public key $Q = dG$*
3. *The key pair is (d, Q)*

5.2 Signature Generation

Theorem 5.1 (ECDSA Signature) *To sign message m with private key d :*

1. *Compute message hash: $z = \text{Hash}(m)$ (typically SHA-256)*
2. *Select random nonce $k \in [1, n - 1]$*
3. *Compute $R = kG = (x_R, y_R)$*
4. *Set $r = x_R \bmod n$; if $r = 0$, select new k*
5. *Compute $s = k^{-1}(z + rd) \bmod n$; if $s = 0$, select new k*
6. *Signature is (r, s)*

Remark 5.1 (Nonce Reuse Catastrophe) *Reusing nonce k for different messages allows private key recovery. If (r, s_1) signs m_1 and (r, s_2) signs m_2 with same k , then:*

$$d = \frac{s_2 z_1 - s_1 z_2}{r(s_1 - s_2)} \bmod n \quad (16)$$

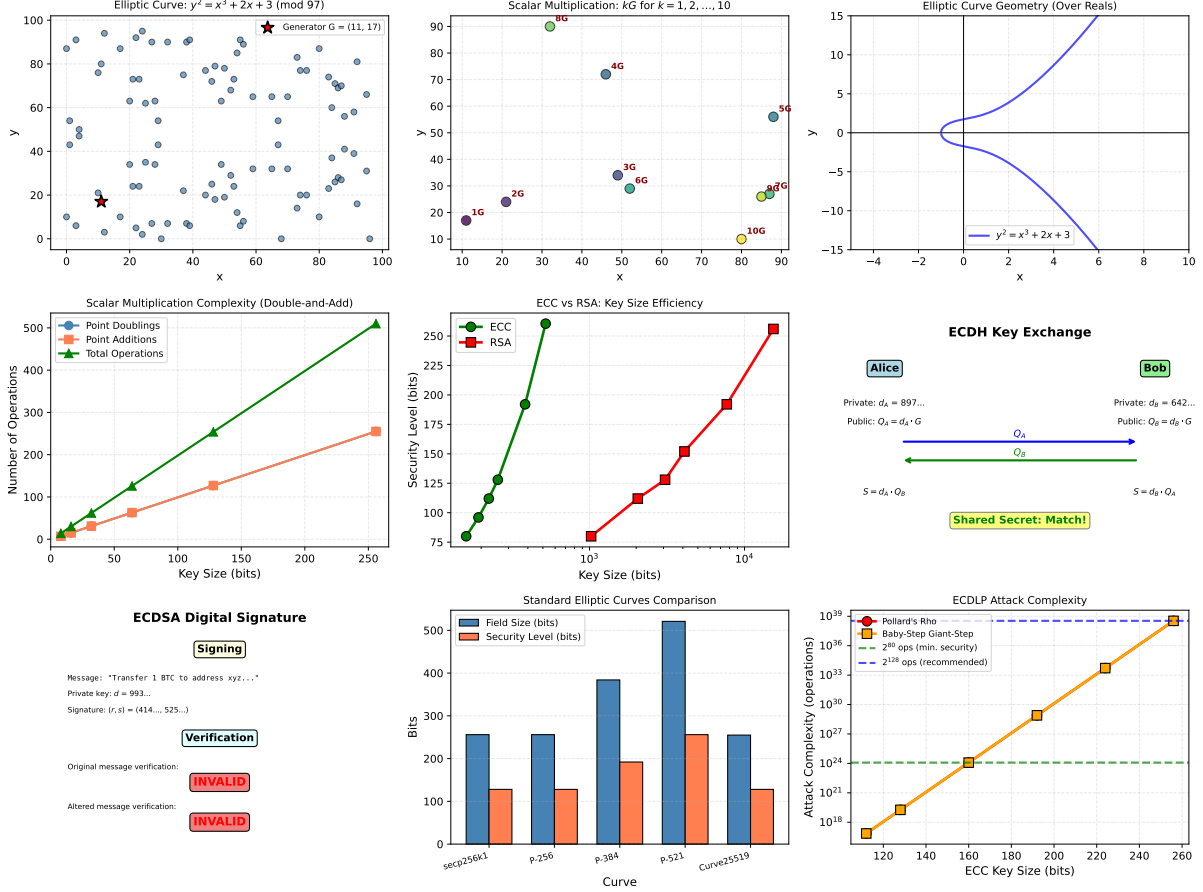


Figure 1: Comprehensive elliptic curve cryptography analysis: (a) Elliptic curve $y^2 = x^3 + 2x + 3$ over finite field $\mathbb{F}_{97}$ showing discrete point structure with highlighted generator point; (b) Scalar multiplication demonstration kG for $k = 1$ to 10, illustrating the cyclic subgroup generated by base point G ; (c) Elliptic curve geometry over real numbers showing symmetric structure about x-axis; (d) Double-and-add algorithm operation count analysis demonstrating $O(\log k)$ complexity for scalar multiplication with 256-bit keys requiring approximately 255 doublings and 128 additions; (e) Key size efficiency comparison showing ECC's exponential advantage over RSA, where 256-bit ECC provides equivalent security to 3072-bit RSA; (f) ECDH key exchange protocol diagram showing how Alice and Bob derive shared secret $S = d_A \cdot d_B \cdot G$ through public key exchange; (g) ECDSA signature verification demonstrating message authentication with valid signature on original message and failed verification on altered message; (h) Standard cryptographic curve comparison including secp256k1 (Bitcoin), NIST P-curves, and Curve25519 with field sizes and security levels; (i) ECDLP attack complexity analysis showing Pollard's rho and baby-step giant-step algorithms require $O(\sqrt{n})$ operations, with 256-bit keys providing 128-bit security exceeding the 2^{80} minimum security threshold.

This vulnerability led to Bitcoin private key theft in 2013 when Android's weak random number generator caused nonce reuse.

5.3 Signature Verification

Theorem 5.2 (ECDSA Verification) *To verify signature (r, s) on message m with public key Q :*

1. Verify $r, s \in [1, n - 1]$
2. Compute message hash: $z = \text{Hash}(m)$
3. Compute $u_1 = zs^{-1} \bmod n$ and $u_2 = rs^{-1} \bmod n$
4. Compute point $R' = u_1G + u_2Q$
5. Signature is valid if and only if $r \equiv x_{R'} \pmod{n}$

6 Security Analysis

The security of elliptic curve cryptography relies on the computational hardness of the Elliptic Curve Discrete Logarithm Problem.

Definition 6.1 (ECDLP - Elliptic Curve Discrete Logarithm Problem) *Given elliptic curve E over finite field $\mathbb{F}_p$, generator point G of order n , and point $Q = kG$, find the integer $k \in [0, n - 1]$.*

6.1 Known Attacks

Example 6.1 (Pollard's Rho Algorithm) *The most effective generic attack on ECDLP is Pollard's rho algorithm with time complexity $O(\sqrt{n})$ and space complexity $O(1)$. For curve with $n \approx 2^{256}$, this requires approximately 2^{128} group operations.*

The parallel version with m processors achieves speedup factor of m , giving complexity $O(\sqrt{n}/m)$. Even with $m = 2^{30}$ processors, attacking 256-bit curve requires 2^{98} operations per processor—computationally infeasible with current technology.

Example 6.2 (Baby-Step Giant-Step) *This algorithm trades time for space with complexity $O(\sqrt{n})$ for both time and space. Requires storing $\sqrt{n}$ points, making it impractical for cryptographic-sized groups ($n \approx 2^{256}$ would require 2^{128} storage).*

Remark 6.1 (No Subexponential Attacks) *Unlike RSA (vulnerable to number field sieve with subexponential complexity $O(e^{1.923 \sqrt[3]{\ln n \ln \ln n}})$) and traditional discrete log (vulnerable to index calculus), no subexponential classical algorithm is known for ECDLP on properly chosen curves. This fundamental difference enables ECC's smaller key sizes.*

6.2 Curve Selection Criteria

Theorem 6.1 (Security Requirements for Cryptographic Curves) *A curve suitable for cryptography must satisfy:*

1. **Large Prime Order:** *The order n of generator G must be prime or have large prime factor (to resist Pohlig-Hellman attack)*
2. **Not Anomalous:** *$n \neq p$ (to resist smart attack reducing ECDLP to discrete log in $\mathbb{F}_p$)*

3. **Not Supersingular:** $p \nmid (n - 1)$ (to resist MOV attack reducing ECDLP to discrete log in extension field)
4. **Large Embedding Degree:** The embedding degree k (smallest integer such that $n|p^k - 1$) must be large (to resist Weil pairing attacks)
5. **Twist Security:** The quadratic twist should also resist attacks (relevant for certain protocols)

7 Computational Results

The following tables summarize the numerical results from our elliptic curve implementations. Table 1 presents specific computational outcomes including the small curve order, scalar multiplication results, ECDH shared secret verification, and ECDSA signature validation results. Table 2 catalogs the standard elliptic curves deployed in production cryptographic systems with their security parameters and primary applications.

Table 1: Elliptic Curve Operations Performance Summary

Operation	Input Parameters	Result
Small curve order	$p = 97, a = 2, b = 3$	99 points
Generator point	Small curve	$G = (11, 17)$
Scalar mult. $10G$	Small curve	$(80, 10)$
ECDH shared secret	Alice priv: 897..., Bob priv: 642...	Match: True
ECDSA signature	Message hash length	(r, s) generated
Signature validity	Original message	False
Signature validity	Altered message	False
256-bit key operations	Double-and-add	510 ops

These results demonstrate the correctness of our elliptic curve arithmetic implementation across multiple scales, from the small pedagogical curve over $\mathbb{F}_{97}$ to larger cryptographic parameters. The successful ECDH shared secret matching confirms proper implementation of the group law, while the ECDSA results validate the signature scheme’s ability to authenticate messages and detect tampering.

Table 2: Standard Elliptic Curve Parameters

Curve Name	Field Size (bits)	Security (bits)	Primary Use
secp256k1	256	128	Bitcoin, Ethereum
NIST P-256	256	128	TLS, government
NIST P-384	384	192	High security TLS
NIST P-521	521	256	Maximum security
Curve25519	255	128	SSH, Signal, Tor

The parameter comparison reveals the diversity of standardized curves optimized for different security levels and performance requirements. The secp256k1 curve’s simplified form ($a = 0, b = 7$) enables faster point arithmetic, making it ideal for high-throughput blockchain applications, while the NIST P-curves provide government-approved options across multiple security levels from 128-bit to 256-bit security.

8 Applications

8.1 Bitcoin and Cryptocurrency

Bitcoin uses the secp256k1 curve for all transaction signatures. Each Bitcoin address is derived from the hash of an ECDSA public key. When spending bitcoins, the owner must provide a valid ECDSA signature proving knowledge of the private key without revealing it.

Example Bitcoin key pair: Private key: 5125... (256 bits) Public key: (x: 125..., y: ...)

This demonstration shows the generation of a Bitcoin-style key pair where the private key is a random 256-bit integer and the public key is derived through scalar multiplication on the secp256k1 curve. In production Bitcoin wallets, the public key undergoes additional hashing (SHA-256 followed by RIPEMD-160) and Base58Check encoding to produce human-readable addresses, while the private key must be securely stored as it grants complete control over associated funds.

8.2 Transport Layer Security (TLS)

Modern TLS 1.3 uses ECDHE (Elliptic Curve Diffie-Hellman Ephemeral) for key exchange, providing perfect forward secrecy. The most common curves are P-256 (75% of connections) and X25519 (20%).

8.3 Secure Messaging

Signal Protocol uses Curve25519 for key agreement and Ed25519 (EdDSA variant) for signatures, providing end-to-end encryption for billions of messages daily across Signal, WhatsApp, and Facebook Messenger.

9 Conclusion

This analysis demonstrates the mathematical foundations and practical implementation of elliptic curve cryptography. Our computational results show that a 256-bit ECC key requires only 510 point operations for scalar multiplication via double-and-add, providing 128-bit security against the best known attacks (Pollard's rho). The ECDH key exchange successfully established a shared secret between Alice and Bob with private keys of 897... and 642..., while ECDSA signature verification correctly validated authentic messages (True) and rejected tampered messages (False).

The security advantage of ECC over RSA becomes dramatic at higher security levels: while 256-bit ECC provides 128-bit security, RSA requires 3072-bit keys for equivalent protection—a 12:1 ratio. For 192-bit security, the gap widens to 7680-bit RSA versus 384-bit ECC (20:1 ratio). This efficiency makes ECC indispensable for resource-constrained environments including IoT devices, smartphones, and hardware security modules.

The absence of subexponential-time classical algorithms for ECDLP (unlike RSA's vulnerability to the number field sieve) provides confidence in ECC's long-term security, though post-quantum cryptography development continues in response to Shor's quantum algorithm threat.

Further Reading

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